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The relationship between college students' social media use and core self-evaluation: the chain mediating effect of information overload and fear of missing out

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Abstract

Background Social media has become an integral part of college students' daily lives, yet the issues of information overload and emotional distress it brings are increasingly prominent. Previous studies have primarily focused on the associations between social media use and psychological outcomes such as well-being, anxiety, or depression, while paying insufficient attention to the underlying mechanisms through which it influences individuals' core self-evaluation. This study aims to explore the mediating roles of information overload and fear of missing out (FoMO) in the relationship between social media use and core self-evaluation.

Method A cross-sectional survey was conducted in a prefecture-level city in Henan Province, China, from November to December 2023, involving two universities. The questionnaire was administered online via the Wenjuanxing platform. A total of 1519 college students participated in the cross-sectional survey.

Results Results from the chained mediation model indicate that social media use does not directly and positively predict college students' core self-evaluation; rather, its influence is primarily exerted indirectly through information overload and fear of missing out. Both variables serve as significant mediators and form a sequential mediating pathway.

Conclusion Information overload and FoMO were identified as partial mediators in the correlation between social media usage and core self-evaluation among college students, highlighting the existence of a chain mediation mechanism linking them. This study elucidates the mechanism through which social media influences college

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students' self-cognition. Theoretically, it advances the understanding of the relationship between social media use and core self-evaluation; practically, it offers insights for universities to develop interventions targeting information management and emotion regulation, thereby supporting students' mental health.

Keywords Social media use, Information overload, Fear of missing out (FoMO), Core self-evaluation, College students

Introduction

As of December 2023, the number of internet users in China surged to 1.092 billion. Among these figures, adolescent internet users constituted approximately 14.7% of the total demographic, while the average weekly online engagement per person stood at 30.8 h [1]. Notably, there has been a marked escalation in the utilization of social media platforms by college students, rendering it an indispensable facet of their daily routines. College students use social media to interact with friends, share their lives, obtain information, and express personal views, making them heavy users of social media [2, 3]. However, the burgeoning reliance on social media among adolescents has engendered a host of psychological and societal challenges. Research indicates that social media use may not only diminish college students' self-esteem [4] and self-efficacy, but also increase the risk of Internet addiction [5, 6], sleep disturbances [7, 8], academic burnout [9], as well as anxiety and depression [10, 11], thereby exerting negative effects on their physical and mental health [12].

Exploring the relationship between social media use and core self-evaluation

Adolescence stands out as a critical juncture for the cultivation and refinement of core self-evaluation, playing a pivotal role in the maturation and advancement of individuals [13]. Among college students, a cohort heavily engaged in social media usage, the correlation between their utilization of social media platforms and core self-evaluation has garnered notable attention. Core self-evaluation encompasses a profound personality framework comprising self-esteem, emotional stability/neuroticism, locus of control, and general self-efficacy, encapsulating individuals' fundamental assessments of their competencies and intrinsic worth [14]. These evaluations shape individuals' perceptions and dispositions towards themselves, wielding substantial sway over their mental well-being and overall quality of life. According to the Core Self-Evaluation Self-Cognition Theory [15], individuals with higher levels of core self-evaluation are inclined towards fostering constructive self-cognitive schemas within their cognitive faculties, demonstrating a propensity to acquire and process favorable stimuli from the environment, while those with lower levels of core self-evaluation exhibit wavering confidence in their problem-solving aptitudes, frequently grappling with feelings of inadequacy in navigating present and forthcoming life circumstances, thereby harboring pessimistic outlooks. Research has shown that core self-evaluation is not only a protective factor against suicidal ideation [16] but also a predictor of academic burnout [9]. Moreover, core self-evaluation emerges as a reliable predictor of individuals' anxiety levels [17, 18], underscoring the imperative of prioritizing the nurturing and enhancement of core self-evaluation among college students.

Social media platforms, alternatively referred to as social networks, serve as multifaceted mediums facilitating identity expression, communal engagement, and various applications, enabling users to interact with online communities through the dissemination, exchange, and creation of information [19], such as Weibo, WeChat, Little Red Book (Xiaohongshu), etc. By extending offline social interactions into the digital realm, social media platforms have the capacity to transmute traditional face-to-face socialization into online manifestations, such as through the sharing of photographic or videographic depictions of social engagements [20]. Erikson [21] emphasizes the pivotal role of interpersonal communication and the social milieu in shaping individual self-evolution. Compared to their adult counterparts, adolescents are more sensitive to their surrounding environment and are more susceptible to self-evaluation and external influences [22]. Empirical evidence suggests a close nexus between core self-evaluation and life events, positing that the formation or modification of core self-evaluation is contingent upon environmental stimuli and the internalization of extrinsic information by individuals [23]. This construct not only exerts a favorable impact on individuals' overall life satisfaction but also assumes a pivotal role in modulating affective states and physiological well-being. Against the backdrop of the pervasive ubiquity of social media, it emerges as a salient environmental determinant impinging upon individual development. College students are particularly active on social media, highlighting their high dependence on this platform. Studies have unveiled a negative correlation between overall social media usage and self-esteem [24–26], indicating that individuals who habitually frequent social media channels tend to evince diminished levels of self-esteem. For example, Sharifian et al. [27]found that discerned a positive association between heightened daily social media engagement and heightened incidences of negative affect and memory lapses, concomitant with a reduction in self-esteem levels. Other scholarly investigations have underscored the capacity of social media engagement

to bolster individuals' connectivity with their communities and the broader external milieu [28–30]. By broadening cognitive horizons, augmenting social capital and competencies, social media utilization furnishes avenues for emotional bolstering and fosters the enhancement of individual psychological well-being. Hence, a discernible correlation emerges between social media utilization and individuals' core self-evaluation, warranting concerted scholarly attention and inquiry into this intricate interplay.

Information overload

The widespread adoption and escalating utilization of social media platforms have precipitated an unprecedented surge in the volume of information to which individuals are exposed. Information overload delineates a scenario wherein the quantity of information surpasses an individual's cognitive processing capacities [31]. This phenomenon pervades contemporary society, characterized by instances wherein individuals find themselves ensconced in social media engagements from dawn till dusk, even perusing assorted information sources prior to retiring for the night. This perpetual influx of information confers palpable stress upon individuals, albeit paradoxically, endeavors to assuage this strain through social media utilization frequently perpetuate a pernicious cycle of information inundation. Present-day society confronts the ubiquitous challenge of heightened frequency and magnitude of information consumption, precipitating a state of information overload across individuals, collectives, and societal strata.

Statistical data reveals that individuals dedicate an average of 2.5 h daily to smartphone usage, with college students dedicating upwards of 5 h, while the mean weekly internet consumption per capita in China stands at 29.1 h [1, 32–34]. Given the relatively homogeneous demographic characteristics of college students in terms of educational attainment, age distribution, and social media engagement patterns [35], the proclivity towards information overload is notably pronounced. Scholarly investigations underscore the pervasiveness of information overload among social media users. A survey conducted by Bontcheva et al. [36] among 587 British Twitter users elucidated that two-thirds of respondents experienced information saturation, with over half advocating the necessity for tools to winnow out extraneous content. Similarly, a study by Sasaki et al. [37] involving 778 Japanese Twitter users revealed that a majority (50.6%) expressed feelings of inundation stemming from the profusion of tweets or information, with 63.7% opining that an excess of vacuous content pervaded the platform. Additionally, Gomez-Rodriguez et al. [38] further confirmed the ubiquity of information overload among Twitter users in Germany through the analysis of data from active users. Therefore, the utilization of social media platforms engenders a propensity towards information overload among individuals.

The body of research underscores a coherent linkage between "information overload - mindset change," delineating a trajectory that exerts profound ramifications on individuals [39]. When recipients exhibit a dearth of requisite background knowledge essential for processing incoming information efficaciously, instances of information overload ensue [40], culminating in the proliferation of negative emotional responses among users [41]. Notably, information overload manifests a symbiotic relationship with core self-evaluation, precipitating adverse consequences such as diminished self-esteem [42] and heightened proclivities towards depression [43]. Individuals ensnared in a protracted state of anxiety arising from information overload [44], grapple with stressors, cognitive dissonance, and self-doubt, thereby jeopardizing their mental and physical well-being [45].

Notably, information overload is closely associated with core self-evaluation and has been linked to reduced self-esteem [42] and an increased risk of depression [43]. Individuals experiencing prolonged anxiety due to excessive information [44] often struggle with heightened stress, confusion, and self-doubt, which can ultimately undermine their mental and physical well-being [45].

Moreover, information overload is closely related to core self-evaluation. Experiencing information overload can lead to reduced self-esteem [42] and depression [43], as individuals often remain in a prolonged state of anxiety [44], feel pressured, struggle to keep up with ongoing events, and may even experience confusion and self-doubt, ultimately affecting their physical and psychological well-being [45].

Soroya et al.'s study [46] showed that information overload stemming from an excess of informational inputs exacerbates feelings of information anxiety. Empirical studies by Wirth et al. [47] confirmed that information overload leads to frustration among social media users. In addition, research by Gao et al. [48] and Chaouali [49] expounded upon the salient impact of information inundation on the emotional exhaustion experienced by users of social networking sites (SNS); while empirical studies by Yu et al. [50] and Cao et al. [51] also corroborated the significant influence of information overload on the emotional exhaustion observed among Chinese social media patrons. Therefore, this scholarly inquiry posits that information overload may serve as a mediating factor in shaping the intricate nexus between social media utilization and core self-evaluation, thus signifying its pivotal role in modulating individuals' cognitive frameworks and psychological well-being.

Fear of missing out (FoMO)

The phenomenon of FoMO is progressively gaining prevalence, particularly within the backdrop of the extensive adoption of social media platforms [52, 53]. FoMO delineates a pervasive apprehension experienced by individuals stemming from the perceived exclusion from others' significant experiences or valuable information [54]. Functioning as a negative emotional catalyst, FoMO stands as a pivotal determinant driving excessive internet engagement among users [55]. Scholars are increasingly focusing on unraveling the ramifications of FoMO on individuals' psychological well-being and behavioral propensities. Evidence underscores the intimate nexus between FoMO and adverse emotional states such as anxiety and depression [56], alongside its confluence with patterns of social media utilization behavior [57]. Therefore, this has become a focal topic in contemporary research.

Against the backdrop of the burgeoning popularity of social media platforms, individuals are increasingly drawn to these digital spaces to glean insights into others' activities and lives, aiming to avert the risk of missing out on enriching experiences. The accessibility and immediacy afforded by social media cater to the emotional and social requisites of college students, albeit concomitantly engendering the quandary of information overload, thereby nurturing sentiments of FoMO. Notably, when individuals disengage from the virtual realm, they may encounter feelings of anxiety, stress, and frustration catalyzed by the apprehension of overlooking crucial information [58, 59]. Research by Hayran et al. [60] showed that 81% of individuals have experienced FoMO, with college students being particularly affected. According to social comparison theory [61], individuals are predisposed to evaluate themselves and their life trajectories through comparative analyses with their counterparts, especially peers. Notably, social media platforms often serve as conduits for the dissemination of idealized representations, particularly prevalent among adolescents [62]. Adolescents who exhibit heightened levels of social media engagement are more inclined towards engaging in social comparisons, thereby fostering a potentiation of FoMO tendencies [63]. Individuals evincing elevated FoMO levels demonstrate a predilection for increased social media usage to satiate their social relational needs when juxtaposed against their counterparts exhibiting lower FoMO proclivities [64, 65]. Consequently, a positive correlation emerges between the frequency of social media utilization and the intensity of an individual's FoMO concerning pivotal information [66].

Previous studies have established a notable inverse correlation between FoMO and self-esteem among social media users [67]. Weaver et al. [68] found that the FoMO among adolescent social media users could influence their self-esteem through a partially mediating effect of online vulnerability. Self-esteem, an overarching assessment of one's self-worth, stands as a pivotal element within core self-evaluation. High levels of FoMO frequently correlate with diminished core self-esteem, as excessive apprehension regarding missed information often signifies doubts or uncertainties concerning one's own capabilities and value. Therefore, this study posits that the utilization of social media may incite FoMO, subsequently intertwining with core self-esteem. In other words, FoMO may serve as a mediating factor in the relationship between social media utilization and core self-esteem.

Previous studies have indicated that information overload can lead to fear of missing out [69]. Specifically, individuals may encounter challenges in effectively assimilating and leveraging copious amounts of information, thereby harboring concerns about overlooking significant content or opportunities. For college students, the utilization of social media to maintain connections with peers, broaden their knowledge networks, and locate resources for information exchange assumes paramount importance in shaping their social identity amongst their peers, particularly in bolstering popularity and fostering a sense of belonging [70]. In today's modern society, individuals have grown accustomed to utilizing social media platforms for accessing diverse forms of information at their convenience. However, this usage pattern has transitioned from a mere desire to stay abreast of others' updates to a psychological state characterized by a FoMO on critical information. The abundance of information and the pervasive push notifications on social media surpass users' finite capacity for information assimilation and processing [71]. Confronted with a vast and intricate volume of data, users often grapple with extracting the pertinent information they require, exacerbating their fear of missing out. Therefore, this study posits that social media usage may induce individuals to confront information overload, which, in turn, may engender Fear of Missing Out, ultimately impacting individuals' core self-evaluation.

Model construction

The core self-evaluation exerts a significant impact on individuals' psychological well-being and life satisfaction. This study aims to investigate the relationship between college students' social media use and their core self-evaluation, as well as the chain mediating effects of information overload and FoMO. Empirical investigations will be undertaken to substantiate our hypotheses and delve deeper into the mediating roles of information overload and FoMO in the correlation between social media consumption and core self-evaluation. This aims to furnish us with a more comprehensive comprehension of the

nexus between college students' engagement with social media platforms and their psychological health, thus offering novel perspectives and avenues for future scholarly inquiry.

This study constructed a chained mediation model (see Fig. 1) to examine the mediating effects of information overload and FoMO on the relationship between social media use and core self-evaluation among Chinese college students. Based on the above discussion, we propose the following hypotheses:

H1 Social media use influences core self-evaluation;

H2 Information overload serves as a mediator in the relationship between social media use and core self-evaluation;

H3 FoMO acts as a mediator in the relationship between social media use and core self-evaluation;

H4 Information overload and FoMO jointly function as mediators in the relationship between social media use and core self-evaluation.

Methods

Participants

This study employed a convenience sampling method to recruit participants. Zhoukou City and Xinyang City in Henan Province were chosen, each housing one comprehensive university, which constituted the focal points of this investigation. Participants completed the online survey questionnaire through the Wenjuanxing platform, the largest online network survey platform in China. Initially, the online questionnaire was disseminated to counselors across various departments of the two universities. These counselors, specialized personnel tasked with overseeing college students, then distributed the questionnaire to class groups. Subsequently, an introductory paragraph outlining the research content preceded the questionnaire. Participation in the survey was contingent upon college students' consent, signified by clicking an agreement button before accessing the online questionnaire.

The inclusion criteria for participants encompassed: (1) enrollment as full-time undergraduate students; (2) proficiency in social media usage; (3) voluntary engagement in the study.

Sample size estimation was conducted employing the Kendall sample estimation method, aiming for a sample size ranging from 10 to 20 times the number of questionnaire items. Given a total of 37 questionnaire items and factoring in a non-response rate of 20%, the minimum required sample size for this survey amounted to at least 888 participants. A total of 1,600 questionnaires were collected in this study. Based on quality control criteria—including abnormal response time, consecutive identical responses, and errors on logic-check items—81 questionnaires were excluded, resulting in 1,519 valid responses.

Measures

Social media use

This study employed the Chinese version of the Social Media Usage Intensity Questionnaire developed by Ellison et al. [72] and revised by Niu et al. [73].

The questionnaire consists of 8 items in total. The first two items assess the number of social media contacts ("How many contacts do you have on social media websites?") and the daily duration of usage ("On average, how much time do you spend on social media?"), with response options spanning from 1 (less than 10) to 10 (over 400) for contacts, and from 1 (less than 10 min) to 10 (over 7 h) for usage duration. The remaining 6 items employ a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree) to gauge the extent of

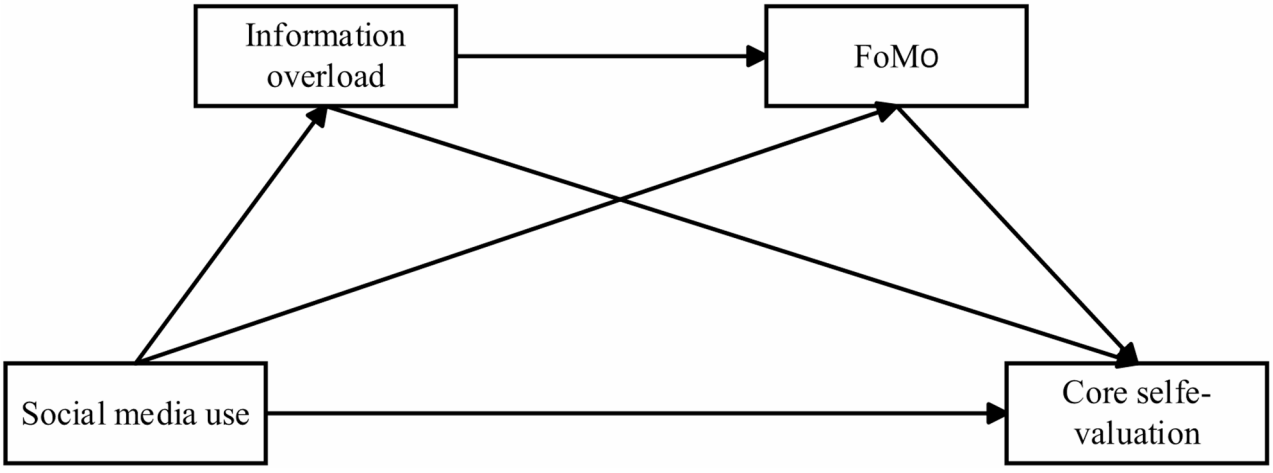


Fig. 1 Presents the proposed sequential mediation model, in which information overload and FoMO serve as mediators between social media use and core self-evaluation

emotional engagement on social media platforms. The scores of each item are aggregated to derive a total score, wherein higher values signify greater usage levels on social media platforms by the individual.

In the study by Liu et al. [74], the scale demonstrated good reliability and validity. In the present study, the Cronbach's α coefficient of the scale was 0.850.

Core self-evaluation

This study used the Chinese version of the Core Self-Evaluation Scale developed by Judge et al. [14]and revised by Ren et al. [75] to measure individuals' cognition and evaluation of their core traits and abilities.

The scale consists of 8 items, with sample items including "Sometimes I feel depressed" and "Sometimes when I fail, I feel worthless." Participants are required to rate each item on a 5-point Likert scale spanning from 1 (strongly disagree) to 5 (strongly agree). The cumulative scores for all items are aggregated to derive a total score, where elevated totals signify increased levels of core self-evaluation by the respondent.

In the study by Liu et al. [74], the scale exhibited favorable reliability and validity. Within the context of this investigation, the Cronbach's α coefficient of the scale is reported to be 0.822.

Information overload

This study used the Chinese version of the Information Overload Scale developed by Koroleva et al. [76], Chen et al. [77], and revised by Zhang et al. [78] to assess individuals' perceived stress and distress levels in relation to information processing.

The scale consists of 4 items, with sample items including "The excessive inflow of information on social platforms like QQ/WeChat/Douyin frequently leads me to feel irritated" and "Every day there is a lot of information to deal with on social software such as QQ/WeChat/Douyin, and I feel overwhelmed." Participantsare required to rate each item on a 5-point Likert scale spanning from 1 (strongly disagree) to 5 (strongly agree). The cumulative scores for each item are aggregated to generate a comprehensive score, where elevated values signify heightened levels of information overload experienced by the individual.

In the study by Xu et al. [79], it was determined that this scale exhibited commendable reliability and validity. Notably, within the confines of the present examination, the reported Cronbach's α coefficient associated with the scale stands at 0.890.

Fear of missing out (FoMO)

This study utilized the Chinese version of the FoMO Scale developed by Przybylski et al. [54] and revised by Wei [80] to primarily measure individuals' level of FoMO.

The scale consists of 10 items, with exemplar statements like "I worry that others have more rewarding experiences than me" and "I worry that my friends have more rewarding experiences than me." Participants are required to rate each item on a 5-point Likert scale, ranging from 1 (completely disagree) to 5 (completely agree). The cumulative scores of all items are totaled to derive an overall score, where higher values indicate greater levels of FoMO experienced by the individual.

In the study by Brailovskaia et al. [81], the reliability and validity of this scale were found to be satisfactory. In this study, the Cronbach's α coefficient of the scale stands at 0.914.

Quality control

The online survey questionnaire released on the Wenjuanxing platform can only be successfully submitted after answering all questions, enforcing a singular submission per mobile device. Therefore, there are no instances of missing data within the compiled questionnaire set, mitigating the likelihood of duplicate submissions. Facilitating streamlined data collection, Wenjuanxing allows for direct exportation of questionnaire data into Excel text format, obviating the need for manual data entry and effectively circumventing potential inaccuracies. To uphold data integrity and quality, all questionnaires undergo meticulous screening in two sequential stages. Firstly, responses completed in under 2 min are excluded, leveraging insights provided by the Wenjuanxing platform. Subsequently, questionnaires featuring identical responses are systematically expunged from consideration.

Ethical considerations

This study adheres to the principles outlined in the Helsinki Declaration andhas garnered ethical endorsement from the Medical Ethics Committee of Xinyang Normal University (Approval No.: XFEC-2024-002). The participants were undergraduate students who were fully informed about the study and voluntarily agreed to take part. According to relevant ethical guidelines, college students aged 17 years and above are regarded as having independent civil capacity and are therefore able to provide informed consent on their own. Consequently, this study did not require additional consent from the guardians of participants under the age of 18. Prior to commencing the electronic questionnaire, participants are briefed on the study's objectives. Only upon consenting, as indicated by clicking the informed consent button, are respondents permitted to proceed with completing the survey questionnaire. All information shared by participants is handled with utmost confidentiality.

Data analysis

This study used SPSS 19.0 (SPSS; IBM, Armonk, NY, USA) and AMOS 28.0 (SPSS; IBM, Armonk, NY, USA) for data analysis. To mitigate common method bias in the research findings, the Harman single-factor test was employed within the SPSS 19.0 statistical software to assess any potential bias. Only after confirming the absence of common method bias, subsequent statistical analyses were conducted. Descriptive statistical analysis, independent samples t-tests, and analysis of variance were carried out using SPSS 19.0 software. Pearson correlation analysis followed in SPSS 19.0 to explore the relationships among variables. Furthermore, AMOS 28.0 software was utilized to investigate the mediating effects of information overload and FoMO on the association between social media use and core self-evaluation. The Bootstrap method was integrated with 2000 bootstrapping samples to analyze 95% confidence intervals, thereby assessing the mediating effects. Lastly, a serial mediation model was formulated using AMOS 28.0 software. In this model, social media use was considered the independent variable, core self-evaluation as the dependent variable, and information overload along with FoMO as mediating variables.

Results

Common method deviation

The Harman single-factor test was used to conduct exploratory factor analysis on all variables. The results revealed six factors with eigenvalues greater than 1, and the first common factor explained 35.2% of the variance, which is less than the critical criterion of 40% [82]. This indicates that there is no significant common method bias in this study.

Collinearity diagnostics were conducted for the independent variables in the study, and the results are presented in Table 1. The tolerance values for social media use, information overload, and FoMO all exceeded 0.2 and were below 1, the VIF values were all less than 5, and the condition indices were all below 30 (strict criteria). These results indicate that no multicollinearity existed among the three independent variables; therefore, the model constructed with these variables is relatively stable.

Demographic information

A total of 1519 college students participated in this study (response rate: 94.94%). Among them, there were 229 (15.1%) males and 1290 (84.9%) females. Regarding the duration of social media use, 44% reported a social media usage span exceeding 5 years, 14% within the 4–5 year bracket, 17.4% spanning 3–4 years, and 24.6% using social platforms for less than 3 years. In terms of weekly usage, 31.5% allocated more than 7 h to social media, 11.3% between 5 and 7 h, 21% spent 3–5 h, 26.2% dedicated 1–3 h, while 10% utilized social media for less than an hour each week. Comprehensive details are elucidated in Table 2.

Table 1 Multicollinearity diagnostics

Social media use	Variable	Condition Index	VIF
	14.528	**0.850**	**1.176**
Information overload	10.049	0.789	1.268
FoMO	13.275	0.700	1.428

The study's findings revealed statistically significant variances in core self-evaluation ($p<0.05$), information overload ($p<0.05$), and FoMO ($p<0.05$) across distinct age categories. Moreover, within real-life scenarios, the quantity of close companions exhibited notable differences concerning social media utilization ($p<0.05$), core self-evaluation ($p<0.05$), information overload ($p<0.05$), and FoMO ($p<0.05$). Additionally, remarkable distinctions were identified in core self-evaluation ($p<0.05$) and information overload ($p<0.05$) among divergent personality profiles. Furthermore, significant variations in core self-evaluation ($p<0.05$) were observed among different academic grade levels.

Descriptive and correlational analyses

Descriptive statistics and Pearson correlation analyses were conducted on social media use, core self-evaluation, information overload, and FoMO. The outcomes unveiled a weak positive correlation between social media use and core self-evaluation, information overload, and FoMO. Notably, information overload displayed a moderate positive correlation with FoMO and core self-evaluation. Furthermore, core self-evaluation exhibited a strong positive correlation with FoMO. Across these relationships, correlation coefficients ranged from 0.203 to 0.662. For an extensive breakdown of these results, please refer to Table 3.

Mediation effects of social media use, core self-evaluation, information overload, and FoMO

As shown in Table 4; Fig. 2, the model reveals that social media use exerts a noteworthy positive predictive influence on information overload ($\beta=0.203$, $p<0.001$). Moreover, information overload demonstrates a significant positive predictive impact on core self-evaluation ($\beta=0.244$, $p<0.001$). Additionally, social media use exhibits a substantial positive predictive effect on FoMO ($\beta=0.305$, $p<0.001$), with FoMO further influencing core self-evaluation positively ($\beta=0.535$, $p<0.001$). Notably, information overload plays a crucial role by significantly prognosticating FoMO ($\beta=0.397$, $p<0.001$). Social media use does not present a significant positive predictive effect on core self-evaluation ($\beta=0.039$, $p>0.001$). This implies that social media use indirectly forecasts college

Table 2 Distribution of participant demographics($N = 1519$)

Variable	Category	N	Social media use	Self-evaluation	Information overload	FoMO
			M ± SD	M ± SD	M ± SD	M ± SD
Gender	Male	229	19.93 ± 4.62	24.89 ± 5.22	12.52 ± 3.27	30.28 ± 6.89
	Female	1290	19.96 ± 4.02	24.44 ± 4.14	12.22 ± 2.87	29.50 ± 6.10
	p		0.092	0.224	0.148	0.082
	t		-0.101	1.219	1.447	1.740
Age	<18	114	20.38 ± 4.25	4.02 ± 0.38	3.04 ± 0.29	5.89 ± 0.55
	18	1134	19.91 ± 4.01	4.17 ± 0.12	2.85 ± 0.09	5.99 ± 0.18
	>18	271	19.99 ± 4.49	4.96 ± 0.30	3.21 ± 0.20	7.25 ± 0.44
	p		0.509	0.008	0.013	0.112
	F		0.676	4.808	4.358	2.192
Grade	Lower grade	1479	19.96 ± 4.12	24.45 ± 4.32	12.26 ± 2.94	29.57 ± 6.25
	Higher grade	40	19.80 ± 4.03	26.73 ± 3.92	12.65 ± 3.01	31.38 ± 5.27
	p		0.806	0.001	0.401	0.071
	t		0.256	-3.297	-0.840	-1.807
Place origin	City	229	20.28 ± 4.76	24.18 ± 4.72	12.20 ± 3.15	29.74 ± 6.68
	County town	341	20.19 ± 3.73	24.35 ± 4.39	12.13 ± 2.91	29.71 ± 6.24
	Rural area	949	19.80 ± 4.08	24.65 ± 4.19	12.33 ± 2. 90	29.56 ± 6.12
	p		0.142	0.246	0.510	0.884
	F		1.956	1.402	0.675	0.123
Nature of subject	Humanities and social sciences majors	868	20.00 ± 4.23	24.36 ± 4.44	12.19 ± 3.01	29.39 ± 6.44
	Natural sciences majors	651	19.91 ± 3.97	24.71 ± 4.16	12.37 ± 2.83	29.93 ± 5.94
	p		0.673	0.115	0.221	0.093
	t		0.423	-1.576	-1.224	-1.682
Monthly living expenses	500–1000	356	19.43 ± 4.28	24.68 ± 4.56	12.23 ± 3.16	29.45 ± 6.50
	1000–1500	866	19.80 ± 3.90	24.46 ± 4.11	12.30 ± 2.85	29.53 ± 6.09
	1500–2000	253	20.93 ± 4.34	24.61 ± 4.40	12.28 ± 2.85	30.04 ± 6.14
	>2000	44	21.68 ± 4.38	23.55 ± 5.72	11.77 ± 3.25	30.27 ± 7.42
	p		0.000	0.399	0.697	0.648
	F		9.785	0.984	0.408	0.706
Number of close friends in real life	0	33	17.42 ± 7.02	24.67 ± 7.32	11.55 ± 4.66	29.39 ± 10.15
	1	80	20.51 ± 3.78	25.65 ± 4.67	13.38 ± 3.70	30.81 ± 6.48
	2	194	19.93 ± 4.05	24.86 ± 4.19	12.58 ± 2.70	30.19 ± 6.08
	3	248	20.04 ± 3.96	25.00 ± 4.42	12.63 ± 2.76	30.40 ± 6.02
	≥ 4	964	19.98 ± 4.05	24.21 ± 4.13	12.04 ± 2.87	29.21 ± 6.10
	p		0.007	0.005	0.000	0.013
	F		3.545	3.690	6.303	3.166
Personality	Introversion	322	20.10 ± 4.34	25.64 ± 4.48	12.89 ± 3.26	30.34 ± 6.79
	Extroversion	262	20.34 ± 4.49	23.43 ± 4.25	11.89 ± 2.83	29.24 ± 6.30
	Both	935	19.80 ± 3.92	24.42 ± 4.20	12.15 ± 2.82	29.48 ± 6.00
	p		0.139	0.000	0.000	0.056
	F		1.974	19.839	10.334	2.891

Table 3 Descriptive statistics and intercorrelations between variables

Variable	1	2	3	4
1. Social media use	1			
2. Core self-evaluation	0.295**	1		
3. Information overload	0.203**	0.497**	1	
4. FoMO	0.386**	0.662**	0.459**	1

Note. ** $p<0.01$

students' core self-evaluation through a sequential mediation pathway involving information overload and FoMO.

From the insights provided in Table 5 concerning the serial mediation model, it is observed that the direct effect of college students' social media use on core self-evaluation stands at 0.039, accompanied by a 95% Boot CI interval of [-0.013, 0.092]. Notably, the total indirect effect value amounts to 0.256, with a 95% Boot CI range of [0.208, 0.302], constituting 86.78% of the

Table 4 The direct effects test of social media use, core self-evaluation, information overload, and FoMO

Path	B	β	SE	t	*p*
Social media use→Information overload	0.145	0.203	0.018	8.078	<0.001
Information overload→Core self-evaluation	0.359	0.244	0.030	11.794	<0.001
Social media use→FoMO	0.462	0.305	0.033	13.922	<0.001
FoMO→Core self-evaluation	0.371	0.535	0.015	24.385	<0.001
Information overload→FoMO	0.842	0.397	0.047	18.080	<0.001
Social media use→Core self-evaluation	0.041	0.039	0.021	1.936	0.053

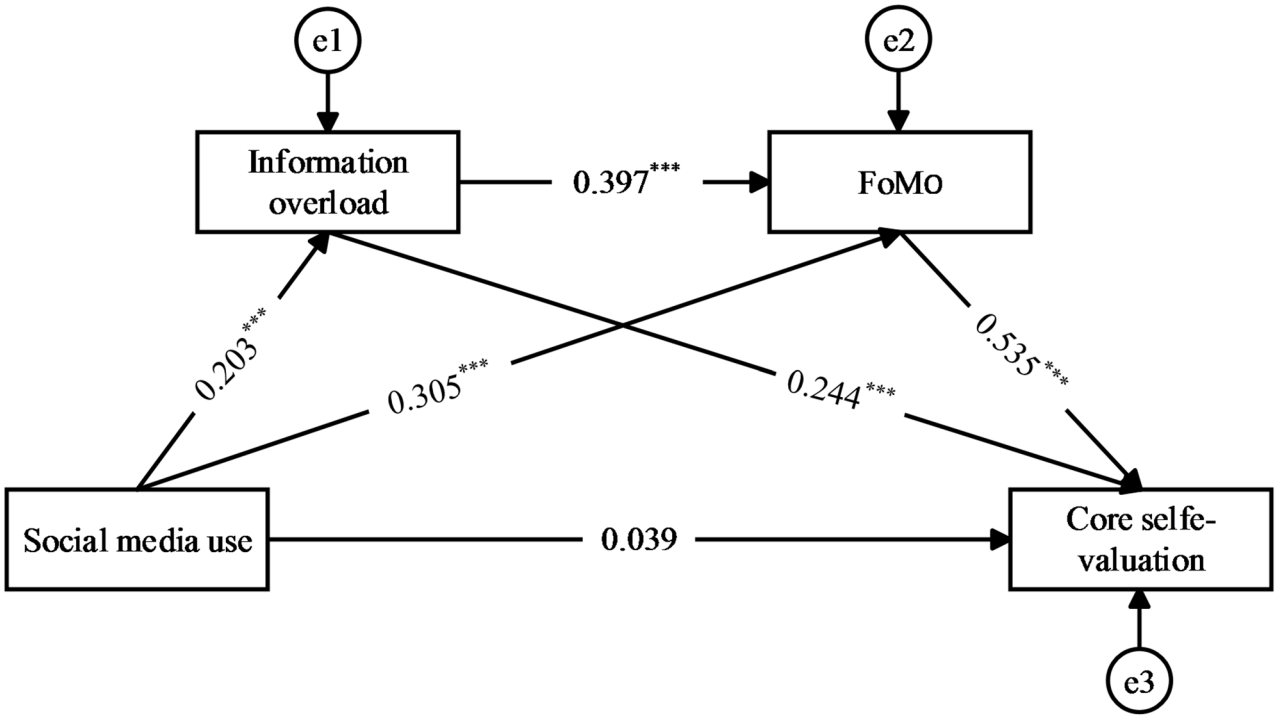


Fig. 2 The indirect prediction of social media use on college students' core self-evaluation: The chain mediating path diagram of information overload and fear of missing out, *** $p < 0.01$

Table 5 Bootstrap analysis of the mediation effect between social media use and core self-evaluation

Model pathways	β	SE	95% Boot CI		VAF[a](%)	*P*
			LL	UL		
Direct effect	0.039	0.027	-0.013	0.092	13.22%	0.001
Total indirect effect	0.256	0.04	0.208	0.302	86.78%	0.001
Social media use→Information overload→Core self-evaluation	0.052	0.011	0.033	0.077	17.63%	0.010
Social media use→FoMO→Core self-evaluation	0.172	0.019	0.134	0.211	58.30%	0.001
Social media use→Information overload→FoMO→Core self-evaluation	0.045	0.010	0.027	0.068	15.25%	0.001
Total effect	0.295	0.039	0.225	0.362	100%	0.001

Note: [a]VAF: These percentages are derived from the squared standardized estimates; VAF[a](%) = (β / Total effect) * 100%, The proportion of each mediating path was calculated by dividing the effect of that path by the total effect; This study did not control for confounding factors

overall effect. Moreover, the path involving social media use-information overload-core self manifests a significant standardized coefficient ($p < 0.01$), presenting a 95% Boot CI within [0.033, 0.077] and an effect size of 17.63%. Similarly, the pathway from social media use to FoMO to core self-evaluation exhibits significance ($p < 0.001$), with a standardized coefficient and a 95% Boot CI encompassing [0.134, 0.211], corresponding to an effect size of 58.30%. Additionally, the route spanning social media use-information overload-FoMO-core self-evaluation emerges as statistically significant ($p < 0.001$) with a standardized coefficient of 0.045, accompanied by a 95% Boot CI range of [0.027, 0.068] and an effect size of 15.25%. In essence, the serial mediation effects orchestrated by information overload and FoMO operating between social media use and core self-evaluation are

Table 6 Structural model assessment and their corresponding result

Hypothesis	β	*p*	Result
Social media use→Information overload	0.203	<0.001	Support
Information overload→Core self-evaluation	0.244	<0.001	Support
Social media use→FoMO	0.305	<0.001	Support
FoMO→Core self-evaluation	0.535	<0.001	Support
Information overload→FoMO	0.397	<0.001	Support
Social media use→Core self-evaluation	0.039	0.053	Not Support

Note: "Supported" indicates that the standardized path coefficient is statistically significant ($p < 0.05$), suggesting that the hypothesized relationship is supported. "Not supported" indicates a nonsignificant path ($p \geq 0.05$)

well-established, underscoring a substantial total indirect effect size of 86.78%.

A comparison of the effect sizes across the three indirect pathways revealed that the indirect effect of social media use on core self-evaluation through FoMO (58.30%) was substantially larger than the indirect effect through information overload (17.63%) and the sequential mediation pathway (15.25%). These findings indicate that FoMO functions as the most influential mediating mechanism in the association between social media use and core self-evaluation.

From Table 6, it can be seen that there is a significant positive predictive relationship between social media use and college students' information overload and FoMO. Both information overload and FoMO have demonstrable capacities to anticipate the core self-evaluation of college students. Notably, the structural equation model ascertains that social media utilization indirectly influences core self-evaluation through both information overload and FoMO as mediating variables. Thus, hypotheses H2, H3, and H4 are substantiated within this research framework. However, although social media use is correlated with core self-evaluation, it does not significantly predict college students' core self-evaluation after controlling for the mediating variables. Therefore, Hypothesis H1 is not supported. This result suggests that the influence of social media use on core self-evaluation is primarily indirect, operating through a sequential chain of mediating variables.

Discussion

This study aims to explore the relationship between college students' social media use and their core self-evaluation and reveal the serial mediation effect of information overload and FoMO. The outcomes distinctly reveal that an exclusive consideration of social media use falls short in effectively prognosticating college students' core self-evaluation. Instead, the study unveils that the prognostic influence of social media engagement on core self-evaluation transpires through the sequential mediation pathway involving information overload and FoMO. All three indirect pathways reached statistical significance, indicating that information overload and FoMO play critical mediating roles in the relationship between social media use and core self-evaluation. These findings suggest that the influence of social media use on core self-evaluation operates primarily through cognitive stress and affective mechanisms rather than through a direct effect. Specifically, information overload reflects the cognitive burden associated with information processing, whereas FoMO represents an affective response linked to social comparison and belonging-related anxiety. Taken together, these pathways imply that, within social media contexts, college students' core self-evaluation is more susceptible to indirect influences mediated by cognitive and emotional processes.

The relationship between social media use and core self-evaluation

The findings of this study indicate that there is no significant relationship between social media use and university students' core self-evaluation. This suggests that core self-evaluation—a relatively stable individual trait—cannot be adequately explained by social media use alone. Core self-evaluation reflects long-term self-perceptions and personality tendencies, and its formation typically involves deeper psychological structures and developmental processes rather than short-term online behavioral patterns.

Existing literature on core self-evaluation has primarily focused on its mediating or moderating roles, rather than treating it as an outcome directly "predicted" by external behaviors. For instance, prior research has shown that core self-evaluation mediates the association between online social behaviors and life satisfaction [83], as well as the relationships between mental health and mobile phone dependence [84], and between depression and suicidal ideation [16]. Moreover, core self-evaluation is identified as an antecedent psychological characteristic of social networking site addiction among Chinese adolescents [85], and low core self-evaluation is significantly associated with cyberbullying-related behaviors [86]. Collectively, these findings suggest that core self-evaluation is more likely to function as a deep-level psychological mechanism that helps explain individuals' behaviors and emotional responses, rather than

being determined by momentary levels of social media use. Therefore, the results of this study do not imply that social media use is entirely unrelated to college students' self-cognition; rather, they highlight that, within a cross-sectional research design, the relationship between social media use and core self-evaluation may manifest as indirect or context-dependent, rather than as a direct linear association.

The mediating role of information overload

The results of this study show that information overload mediates the relationship between social media use and core self-evaluation, thereby supporting Hypothesis H2. Specifically, higher levels of social media use are often accompanied by stronger experiences of information overload, and such experiences are associated with lower core self-evaluation.

Existing research indicates that the abundance, diversity, and rapid updating of information in social media environments are typically associated with higher levels of information overload [78, 87–89]. This line of research focuses on individuals' subjective experiences when confronted with information that exceeds their processing capacity, which is consistent with the findings of the present study. Moreover, previous studies have also reported associations between information overload and various psychological indicators, including reduced self-esteem [42], depressive symptoms [43], and psychological distress [42, 90]. These studies further support the link between information overload and individuals' self-cognitive processes. Meanwhile, several longitudinal studies have examined the relationship between social media use and information overload, finding that YouTube use can directly increase levels of information overload, which in turn is positively associated with depressive symptoms [91]. In other words, when individuals are exposed to large quantities of fragmented and difficult-to-integrate information on social media, they are more likely to experience information overload. Higher levels of information overload subsequently influence individuals' evaluations of their own abilities and worth, indirectly shaping the formation and stability of core self-evaluation. Therefore, information overload may be associated with core self-evaluation to a certain extent.

The mediating role of FoMO

The results of this study show that FoMO mediates the relationship between social media use and core self-evaluation, indicating an association among the three variables and supporting Hypothesis H3. In this study, higher levels of social media use were typically accompanied by stronger FoMO, and FoMO was further associated with lower core self-evaluation.

Existing literature has demonstrated a significant association between social media use and FoMO [92], which aligns with the findings of the present research. In social media contexts, university students tend to pay greater attention to social interactions, group dynamics, and information shared by others. Such a highly comparative social environment may contribute to heightened FoMO experiences. Prior studies have shown that FoMO is associated with various negative emotional states, including psychological discomfort, anxiety, and depression [93, 94]. In addition, FoMO and self-evaluation may influence each other. Existing research has found that FoMO mediates the relationship between social media use and self-esteem [66]. Likewise, studies have identified a link between high levels of COVID-19–related fear and symptoms of smartphone addiction, with psychological resilience potentially mediating this relationship [95]. Furthermore, theoretical and empirical evidence suggests that lower core self-evaluation often co-occurs with more frequent social media use and more pronounced FoMO experiences [85, 96, 97]. Therefore, under varying frequencies and patterns of social media use, university students' FoMO and their core self-evaluation appear to be interconnected to some extent.

The chain mediating effect of information overload and FoMO

This study underscores the pivotal role of information overload and FoMO as serial mediators in the intricate relationship between social media usage and core self-evaluation [69]. This finding is consistent with previous research and provides support for Hypothesis H4. University years represent the period during which mobile social media is used most frequently [98]. Mobile social media serves as an important channel for maintaining social connections [99], providing college students with abundant information and social opportunities. However, such use is often accompanied by information overload, FoMO, and changes in core self-evaluation. First, the constant influx of information, messages, and content on social media platforms places pressure on individuals to process and filter information, leading to experiences of information overload and associated feelings of unease and anxiety. Previous research has shown that information overload can cause individuals to feel confused and anxious [100], which may be related to difficulties in effectively processing information and concerns about missing important content. There is also a certain degree of association between information overload and FoMO. Together, these experiences are linked to individuals' emotional states and mental health, such as increases in negative emotions including anxiety and depression [94], and may be related to changes in the level of core self-evaluation.

Limitations

This study has several limitations. First, because the findings are based on cross-sectional data, causal inferences cannot be drawn. Future research could employ longitudinal designs to further explore the psychological mechanisms underlying college students' social media use. Second, this study adopted convenience sampling and relied on voluntary participation from enrolled college students, constituting a non-probability sampling method that may introduce selection bias and affect the representativeness and generalizability of the findings. Students who voluntarily participated may differ systematically from non-participants in terms of interest, motivation, or patterns of social media use, thereby further limiting the external validity of the conclusions. Future studies may consider using stratified random sampling or recruiting samples from multiple regions to enhance representativeness. Third, this study focused exclusively on college students. Although this population is typical in terms of social media use and self-concept development, the findings primarily apply to this group, and their applicability to individuals of other ages or occupational backgrounds requires further examination. Future research should include more diverse samples to assess the model's applicability across different populations. In addition, although the gender distribution of our sample reflects the typical demographic structure of teacher-training institutions in China, the predominance of female participants may limit the generalizability of the findings. Future studies should incorporate samples with a more balanced gender distribution and examine potential gender differences in these associations. Finally, the data were collected through an online questionnaire platform, which may be subject to various forms of bias. Although efforts were made to minimize potential bias, it may still have affected the accuracy of the results.

Conclusion

This study has significantly contributed to elucidating the intricate mechanisms that interconnect college students' social media usage with their core self-evaluation. By delving into the mediating roles of information overload and FoMO, we aimed to elucidate the underlying associations between social media engagement and core self-evaluation. The outcomes of this investigation indicate that information overload and FoMO serve as partial mediators in the nexus between social media utilization and core self-evaluation, unveiling a consequential serial mediating pathway between these constructs.

These insightful findings hold promise in offering valuable guidance to college students for cultivating healthier social media practices, thereby fostering improvements in their psychological well-being and levels of self-awareness. By acknowledging and comprehending the nuanced relationships uncovered in this study, individuals can potentially navigate their social media interactions more purposefully, contributing to a more balanced and beneficial digital experience.

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Author contributions
LW, WH and LH contributed to the conceptualization, data curation, formal analysis, investigation, methodology, writing of the original manuscript. TW, QJ and ZL contributed to data curation and methodology of the manuscript. QM, SR and MY contributed to the data curation of the manuscript. MP, FW and CW contributed to the methodology, analysis plan, and write of the original manuscript.

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Data availability
Our data were collected for the research group and are not publicly available. he datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Human ethics and consent to participate
This study adheres to the principles outlined in the Helsinki Declaration andhas garnered ethical endorsement from the Medical Ethics Committee of Xinyang Normal University (Approval No.: XFEC-2024-002). Informed consent was obtained from all the participants.

Consent for publication
Not applicable in the declarations section.

Competing interests
The authors declare no competing interests.

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